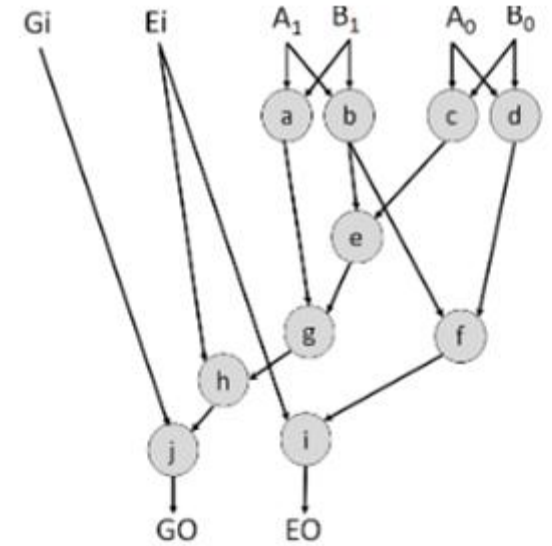


Revision

Q.1

For the Boolean network shown below ($G_i, E_i, A_1, B_1, A_0, B_0$ are Primary Inputs and G_0 and E_0 are Primary Outputs), apply the flow-map algorithm to find the LUT mapping (for $K=3$ i.e. 3-input LUT). Show all the steps. Show the node labeling analysis for each node and the final node mapping (LUT Mapping) for the nodes. Show the final LUT based circuit and in each LUT mention the nodes that the LUT maps.



Q.2

Consider the boolean expressions given below

(a) $F1 = (A \odot B)AB + (A \oplus B)\bar{C}$

(b) $F2 = AB + BC + CA$

For each of the above equations,

- Draw the Ordered BDD directly from equations (without using the truth table) with ordering as $A \rightarrow B \rightarrow C$. Reduce the OBDD to obtain ROBDD
- Based on ROBDD, check if the above Boolean expressions are equivalent.

Assuming that every LUT in the FPGA has 4 inputs and one output. For programming, the LUTs are connected in a shift register. The bit-stream will be shifted in from LUT0 and then pushed to the subsequent LUTs . Mention the bitstream (prepared in the order LUT2→LUT1→LUT0) in hexadecimal. Show calculation [4M]

- (a) **LUT0- 4 input majority function** (Output = 1 when majority of inputs are 1)
- (b) **LUT1- 4 input even parity checker**(Output = 1 when even number of inputs are 1)
- (c) **LUT2- 4 input NAND gate**

LUT0- Majority GATE

A ₃	A ₂	A ₁	A ₀	Output
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

LUT 1 parity generator

A ₃	A ₂	A ₁	A ₀	Count	Output
0	0	0	0	0	1
0	0	0	1	1	0
0	0	1	0	1	0
0	0	1	1	2	1
0	1	0	0	1	0
0	1	0	1	2	1
0	1	1	0	2	1
0	1	1	1	3	0
1	0	0	0	1	0
1	0	0	1	2	1
1	0	1	0	2	1
1	0	1	1	3	0
1	1	0	0	2	1
1	1	0	1	3	0
1	1	1	0	3	0
1	1	1	1	4	1

LUT 2 NAND Gate

A ₃	A ₂	A ₁	A ₀	Output
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

- LUT2: 1111111111111110
- LUT1: 1001011001101001
- LUT0: 0000011001011111

0X F F F E 9 6 6 9 0 6 5 F